

MOSUL, Iraq

WMO: 406080

Lat: 36.306N

Lon: 43.147E

Elev: 709

StdP: 14.32

Time Zone: 3.00 (E03)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 30.4	(c) 33.4	(d) 14.3	(e) 11.7	(f) 43.9	(g) 19.3	(h) 14.9	(i) 54.4	(j) 18.1	(k) 48.6	(l) 15.0	(m) 48.3	(n) 1.8	(o) 340	(p) 0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.4	113.4	69.4	111.4	68.6	109.5	68.3	75.9	101.9	73.0	101.3	71.0	100.8	6.5	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.8	104.9	92.9	63.0	88.4	84.6	60.7	81.4	81.2	39.9	103.3	37.2	101.1	35.3	100.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	13.7	11.9	27.1	117.6	3.1	2.4	24.9	119.3	23.1	120.7	21.3	122.1	19.1	123.8
			25.4	77.4	3.9	6.8	22.6	82.3	20.3	86.3	18.1	90.1	15.2	95.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	71.1	46.3	50.6	57.7	67.4	79.4	90.7	95.7	95.2	85.5	73.9	59.0	49.8
	DBStd	18.32	4.85	5.36	5.89	6.12	6.27	4.38	3.87	3.64	5.17	6.13	5.83	5.44
	HDD50	270	134	52	8	0	0	0	0	0	0	0	4	72
	HDD65	1933	581	403	238	41	1	0	0	0	0	5	193	471
	CDD50	7954	18	70	248	521	912	1221	1417	1401	1065	742	274	65
	CDD65	4141	0	0	12	112	448	771	952	936	615	282	13	0
	CDH74	65866	0	1	152	1252	6006	13043	17017	16344	8882	3034	129	6
	CDH80	44590	0	0	32	464	3393	9052	12617	11974	5612	1435	11	0
Wind	WSAvg	4.2	3.2	3.4	3.8	4.4	5.3	6.0	5.7	5.3	4.3	3.3	2.6	2.7
Precipitation	PrecAvg	18.7	2.3	2.4	2.6	2.7	1.6	0.5	0.5	0.3	0.4	1.3	2.0	2.3
	PrecMax	25.0	3.7	4.5	5.2	4.8	4.7	1.5	1.0	0.7	1.3	3.2	4.7	4.7
	PrecMin	11.2	0.9	0.6	1.1	0.9	0.4	0.1	0.1	0.1	0.0	0.2	0.2	0.5
	PrecStd	3.3	0.8	0.9	1.1	1.1	0.9	0.4	0.2	0.2	0.3	0.8	1.2	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.2	70.1	84.0	98.3	104.4	112.6	116.9	116.6	111.0	98.7	81.1	73.8
		MCWB	51.2	55.4	58.4	62.9	65.6	67.9	69.5	70.6	68.2	64.3	57.5	54.0
	2%	DB	60.4	66.6	77.4	89.8	100.7	109.6	114.5	113.4	106.2	94.9	77.4	66.1
		MCWB	49.7	52.9	56.9	62.8	65.6	68.1	69.7	69.4	66.7	62.8	56.5	51.9
	5%	DB	57.5	64.1	73.8	85.7	98.2	107.6	111.8	111.3	103.1	91.5	74.8	62.3
		MCWB	49.6	51.9	56.1	60.6	64.9	67.2	68.9	68.7	66.1	62.6	56.5	50.5
	10%	DB	55.5	60.8	70.1	81.0	95.0	105.5	109.7	109.3	100.4	88.1	71.2	59.1
		MCWB	48.9	50.6	54.6	59.7	64.1	67.0	68.3	68.5	65.7	61.3	55.5	49.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.4	57.5	62.9	67.3	73.0	83.8	79.8	75.6	72.1	69.3	65.2	56.3
		MCDB	59.3	66.5	73.6	82.5	96.1	103.4	104.7	107.3	99.2	91.3	73.5	64.5
	2%	WB	53.6	55.3	60.2	65.1	69.9	77.6	75.1	73.2	70.0	66.0	62.0	54.5
		MCDB	58.2	63.0	71.0	81.2	91.4	99.0	104.9	103.9	98.2	85.6	70.6	61.5
	5%	WB	51.4	53.5	58.0	63.5	67.9	73.1	72.5	71.6	68.5	64.0	59.8	53.1
		MCDB	55.6	61.7	69.5	78.9	88.7	96.0	103.3	103.6	96.8	83.3	67.8	60.1
	10%	WB	49.7	51.8	56.0	61.9	66.2	69.3	70.3	70.0	67.3	62.8	57.4	51.6
		MCDB	53.6	58.6	67.3	76.6	87.4	97.2	103.5	103.9	95.4	82.8	67.1	57.3
Mean Daily Temperature Range	5% DB	MDBR	17.3	18.0	20.5	23.0	25.1	27.3	27.4	28.4	28.0	25.4	22.1	19.3
		MCDBR	20.7	22.2	26.2	29.5	28.7	29.5	29.6	29.8	30.6	29.2	26.3	23.6
		MCWBR	12.5	11.3	10.9	10.8	9.5	10.1	10.2	10.1	10.7	11.6	12.1	12.6
	5% WB	MCDBR	15.5	18.6	21.7	24.5	26.1	27.7	27.0	29.5	28.5	24.5	19.3	18.3
		MCWBR	10.6	10.2	10.6	11.1	10.4	14.3	11.6	11.9	11.6	11.2	10.6	10.6
Clear-Sky Solar Irradiance	taub		0.404	0.431	0.464	0.543	0.534	0.477	0.495	0.467	0.433	0.468	0.402	0.397
	taud		2.065	2.003	1.941	1.780	1.821	1.948	1.916	1.983	2.063	1.963	2.113	2.096
	Ebn at Noon		238	248	252	239	244	258	252	257	259	232	237	231
	Edh at Noon		40	48	56	69	67	59	61	56	49	49	38	36
All-Sky Solar Radiation	RadAvg		775	1051	1441	1774	2169	2543	2438	2204	1838	1294	948	707
	RadStd		66	106	109	114	124	117	69	76	87	96	89	88

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page